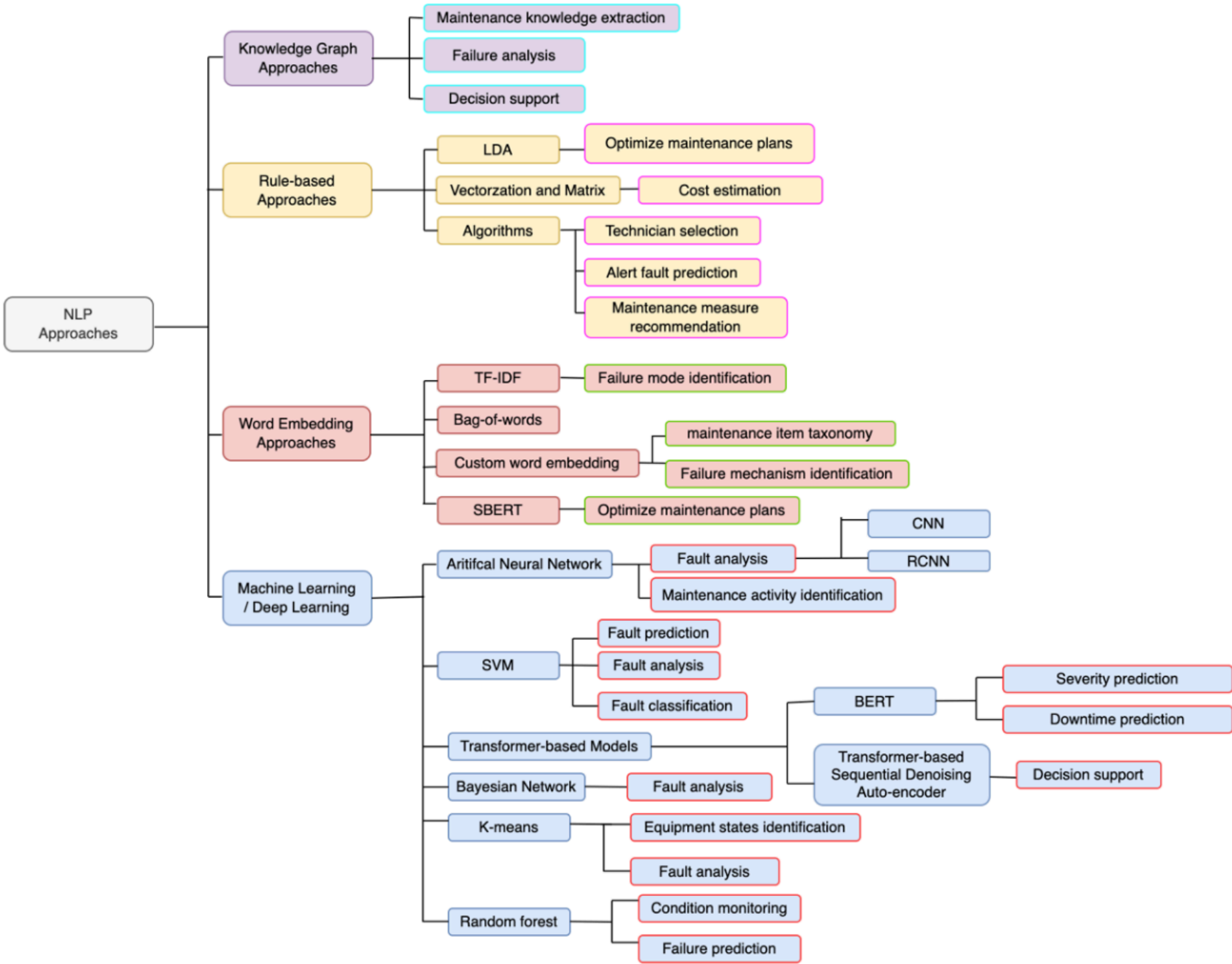




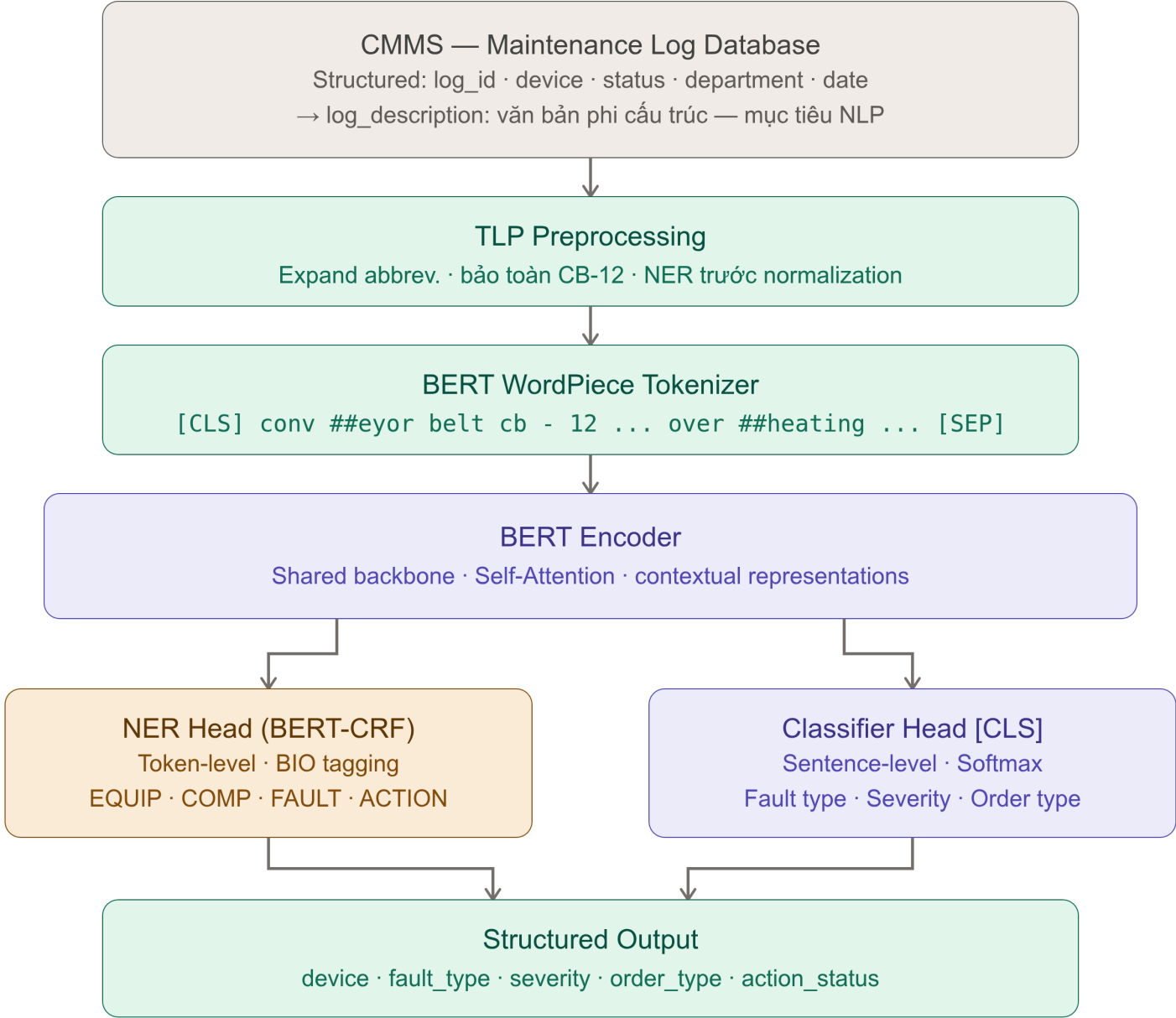
NATURE LANGUAGE PROCESSING CASE STUDY

MAINTENANCE REPORT ANALYSIS

Khái Quát Vấn Đề Thực Tế



Pipeline Mô Hình Maintenance Report Analysis



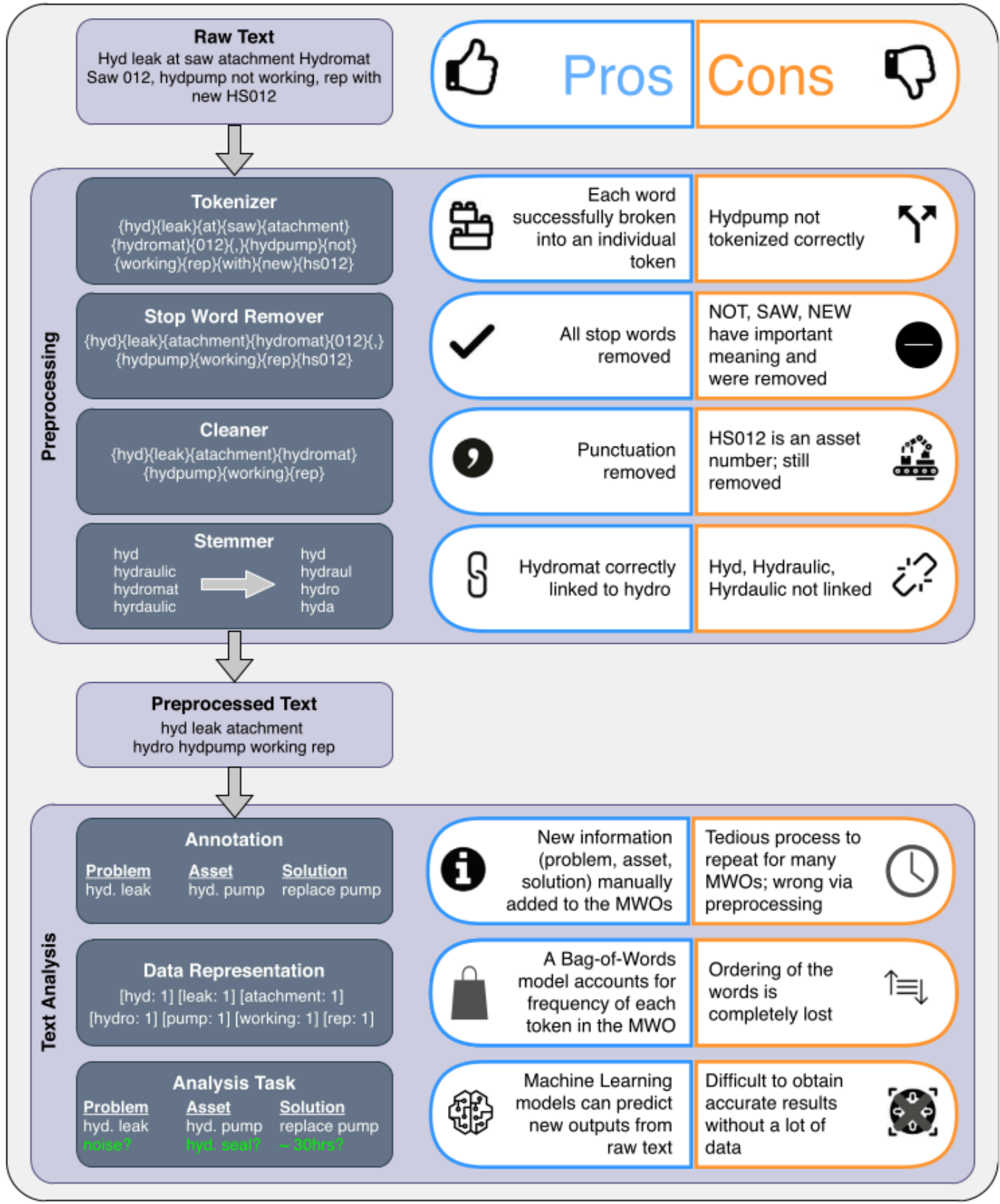
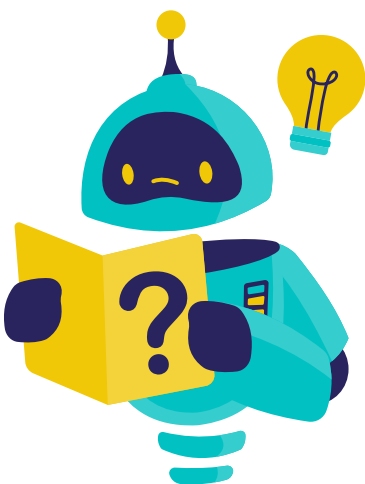
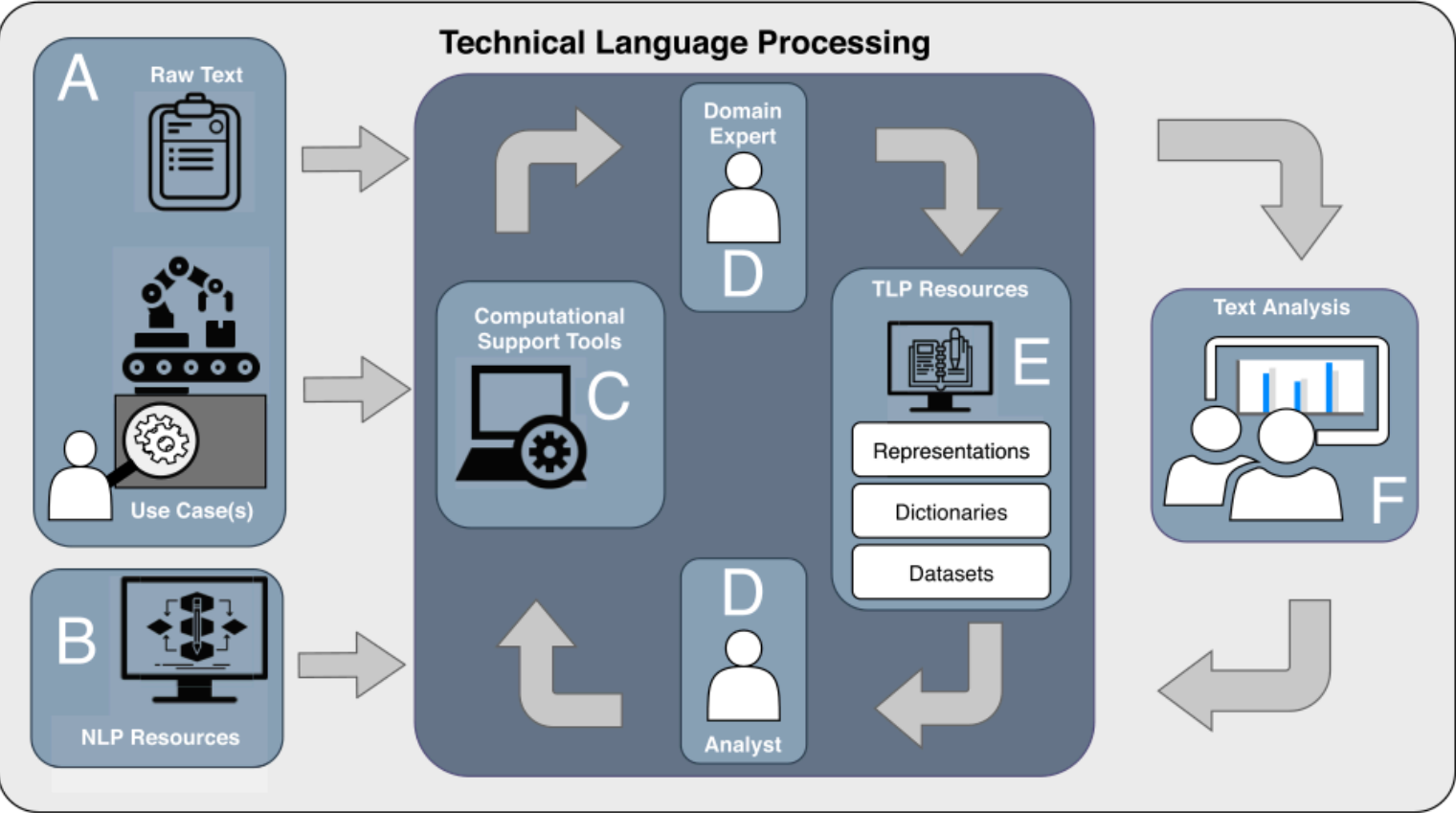
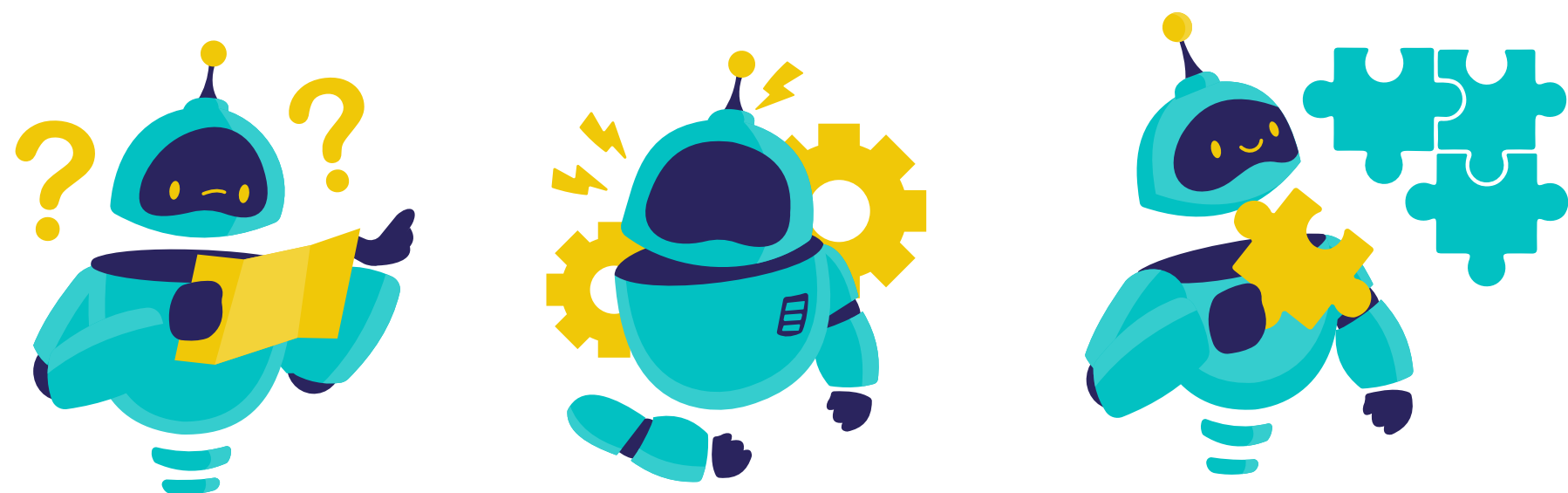
Input - Raw info from CMMS

log_id	log_title	log_description	device	status	emp_name	emp_role	department	date
... [500 data before]								
LOG_00501	Motor Overheating Alert	Conveyor Belt CB-12 stopped unexpectedly due to motor overheating. The motor was replaced and the system was restarted successfully.	CB-12	Solved	William Afton	Maintenance Technician	Mechanical Maintenance	2026-05-15 14:35:00
... [10.000 more]								

Output - Recognition and Classified

log_id	device	severity	fault_type	order_type	action_status
LOG_00501	CB-12	Major	Thermal	Replacement	Solved

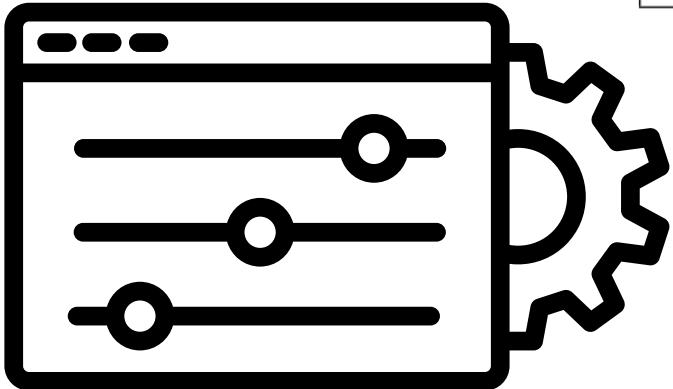
Pre-processing - TLP



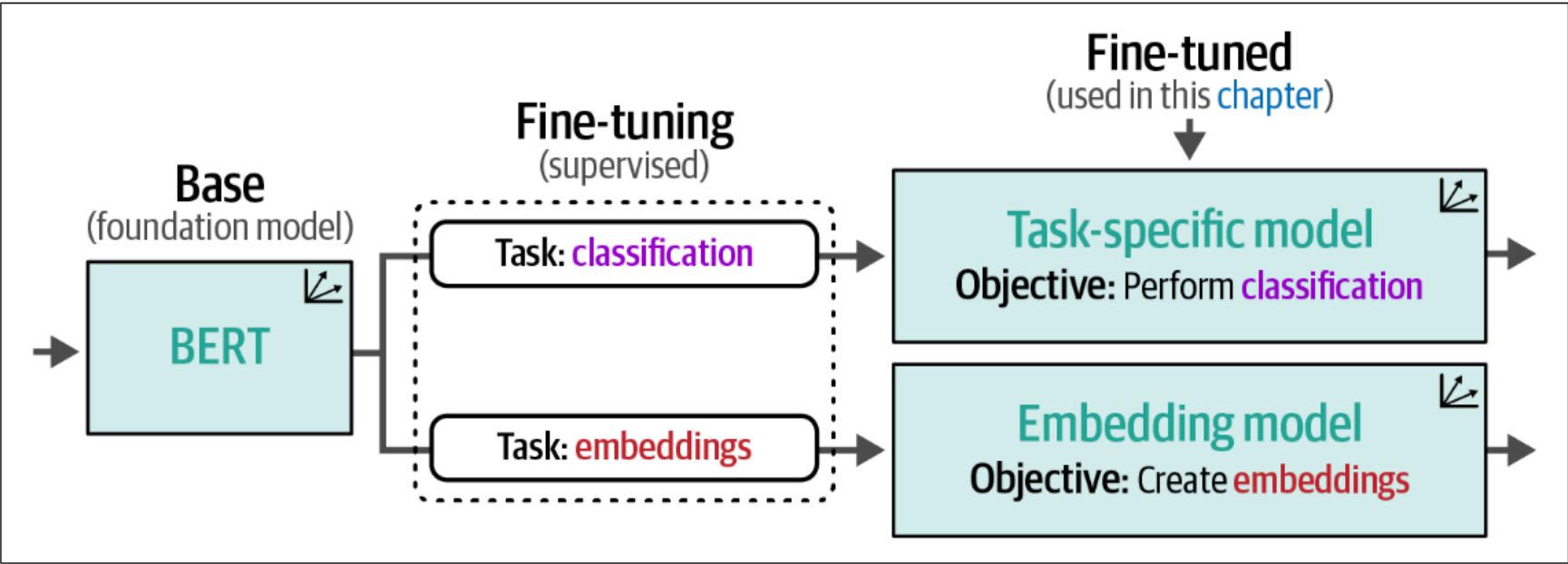
BERT - NER & Text Classification

WordPiece
Subword Tokenize

`overheating`
=> `over` & `##heating`



log_id	device	severity	fault_type	order_type	action_status
LOG_00501	CB-12	Major	Thermal	Replacement	Solved



“Conveyor Belt CB-12 stopped unexpectedly due to motor overheating. The motor was replaced and the system was restarted successfully.”

BERT Token	BIO tag	Entity
[CLS]	O	–
conv	B-EQUIPMENT	EQUIPMENT
##eyor	I-EQUIPMENT	EQUIPMENT
belt	I-EQUIPMENT	EQUIPMENT
cb	I-EQUIPMENT	EQUIPMENT
-	I-EQUIPMENT	EQUIPMENT
12	I-EQUIPMENT	EQUIPMENT
stopped	O	–
unexpected	O	–
##ly	O	–
due	O	–
to	O	–
motor	B-COMPONENT	COMPONENT
over	B-FAULT	FAULT
##heating	I-FAULT	FAULT
.	O	–
the	O	–
motor	B-COMPONENT	COMPONENT
was	O	–
replaced	B-ACTION	ACTION
and	O	–
re	B-ACTION	ACTION
##started	I-ACTION	ACTION
successfully	O	–
[SEP]	O	–

Table 2: Bảng BIO của cột log_description

Đánh Giá

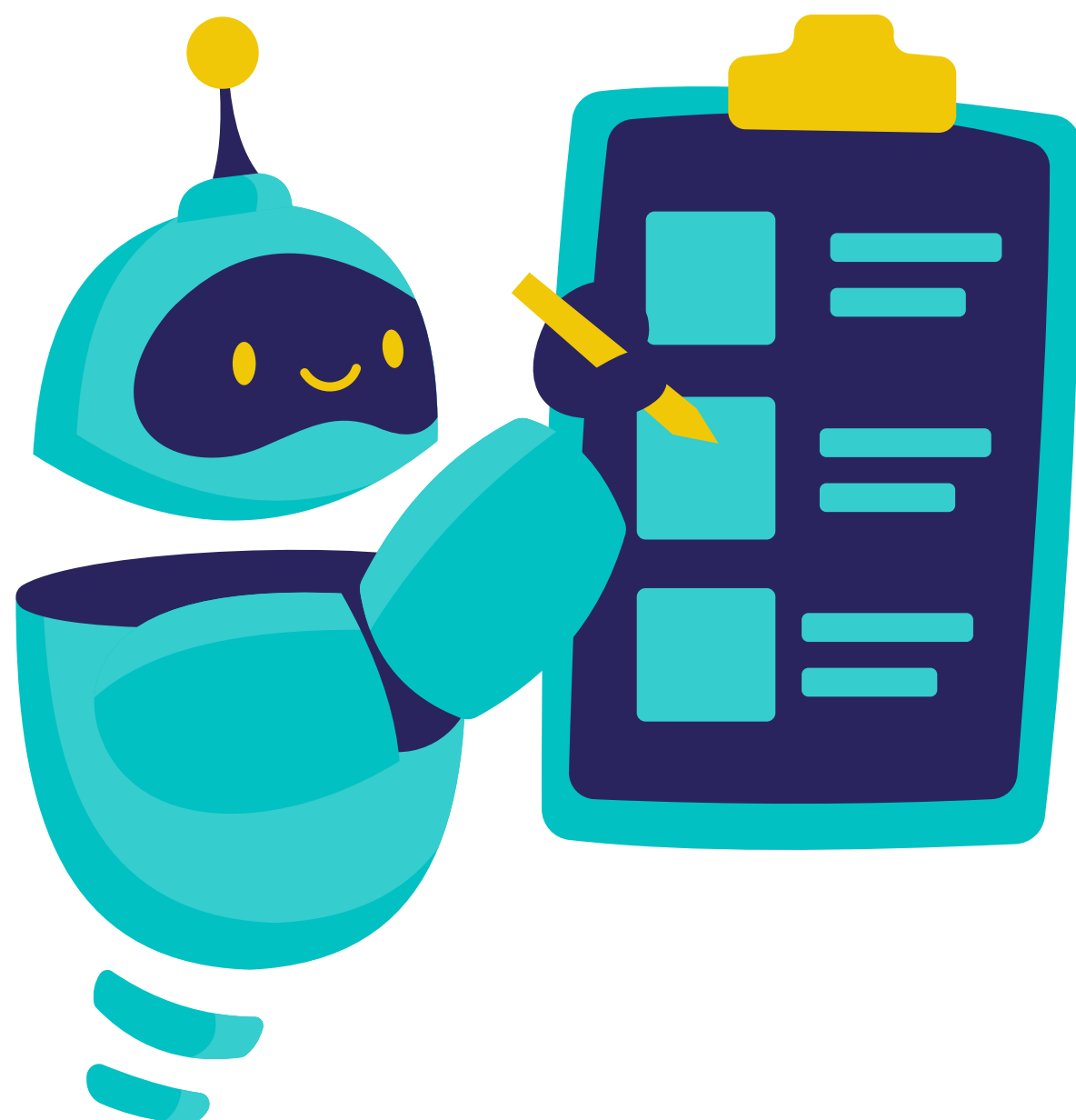


Table 2. Basic information of the three NER datasets

Datasets	Number of words	Number of types	Domain	State-of-the-art (F1 score)	Examples of types
FabNER	350,000+	12	Manufacturing	92%	MATE(Material), PRO(Properties), ENAT(Enabling technology), ...
Thin-film head technology	92,000+	17	Hard-disk	78.2%	Component, Function, PhysicsFlow, EnergyFlow, ...
Assembly NER	22,000+	9	Assembly	84.69%	PART(parts), OPER(operations), TOOL(tools), QTY(Quality),

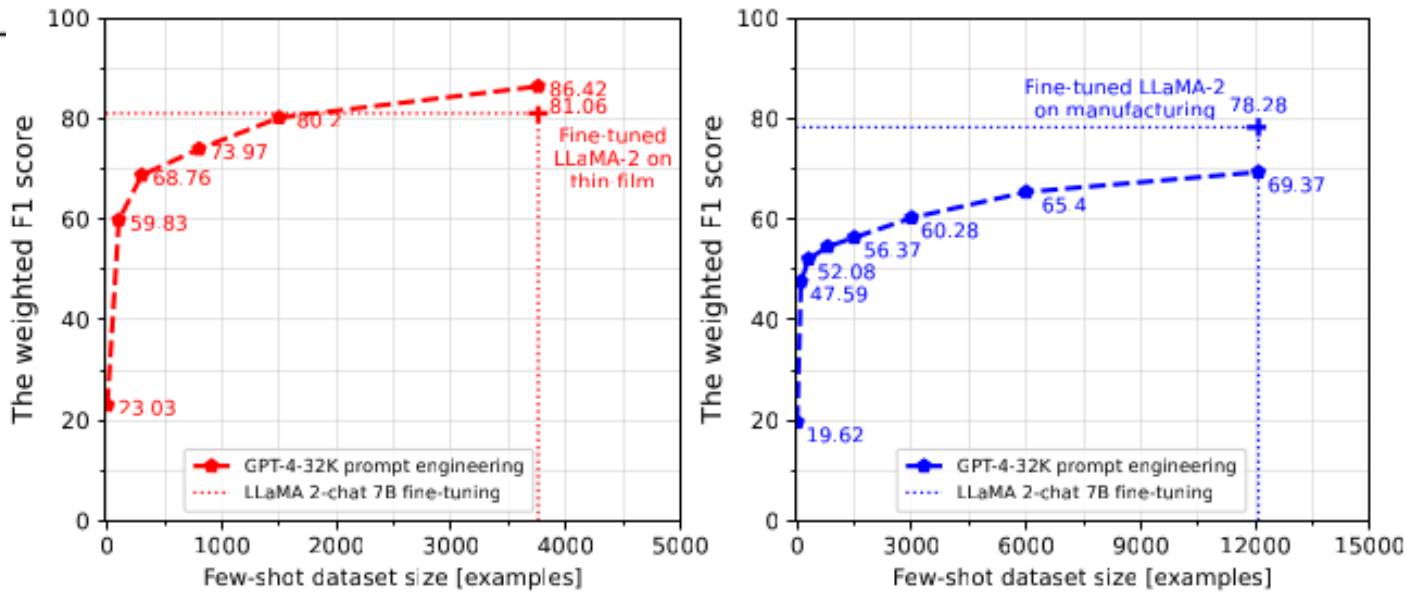


Figure 12. The F1 score of GPT-4 with an increasing size of few-shot dataset in thin-film technology (left) and manufacturing (right) domains.

Thanks for listening

